

ST5209/X Assignment 5

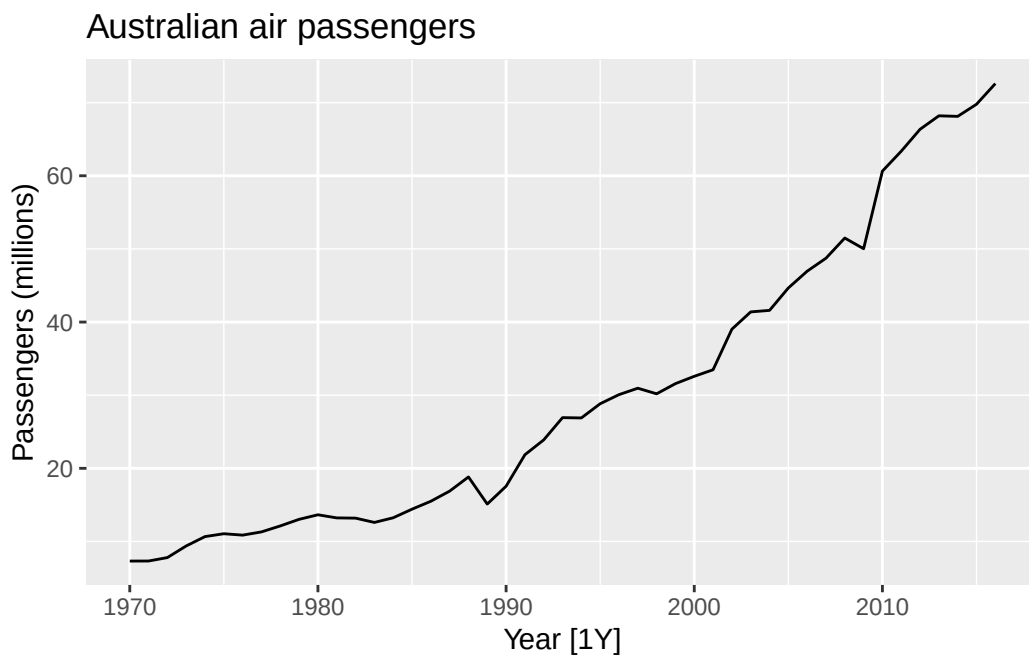
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Submission

1. Render the document to get a .pdf printout.
2. Submit both the .qmd and .pdf files to Canvas.

1. Modeling with ARIMA (Q9.7 in Hyndman & Athanasopoulos)

```
aus_airpassengers |> autoplot(Passengers) +  
  labs(title = "Australian air passengers", y = "Passengers (millions)")
```



- a. Fit an ARIMA model with `ARIMA()`.

```
fit_auto <- aus_airpassengers |>
  model(arima = ARIMA(Passengers ~ 0 + pdq(0, 2, 1)))
report(fit_auto)
```

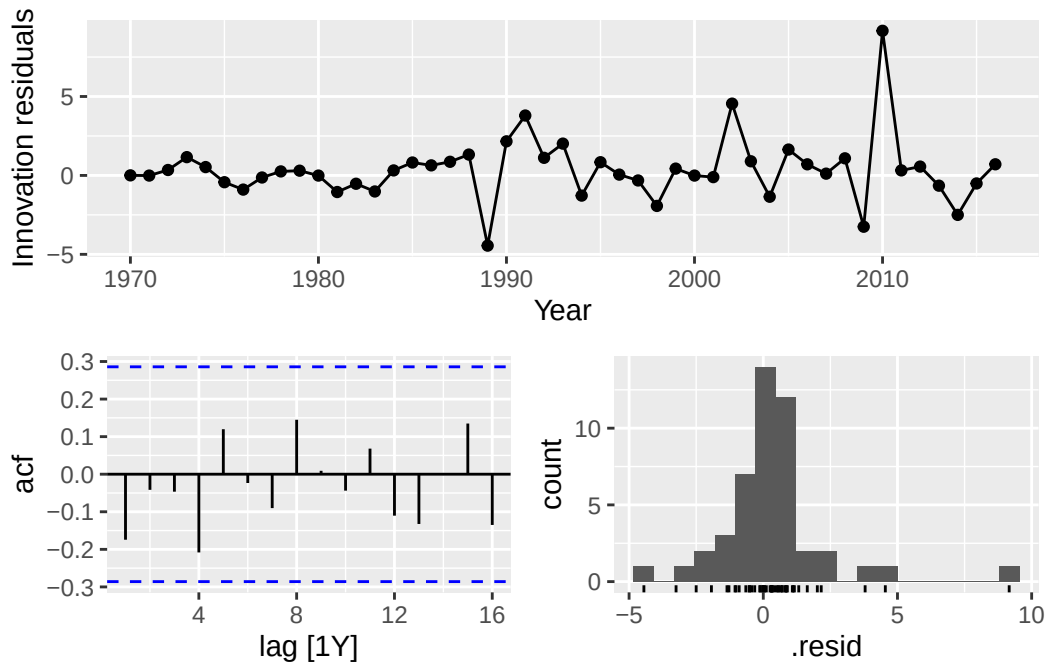
Series: Passengers
Model: ARIMA(0,2,1)

Coefficients:
ma1
-0.8963
s.e. 0.0594

sigma^2 estimated as 4.308: log likelihood=-97.02
AIC=198.04 AICc=198.32 BIC=201.65

Selected: **ARIMA(0, 2, 1)** (no constant), $\hat{\theta}_1 = -0.8963$ (s.e. 0.0594), $\hat{\sigma}^2 = 4.308$, AICc = 198.32.

```
fit_auto |> gg_tsresiduals()
```

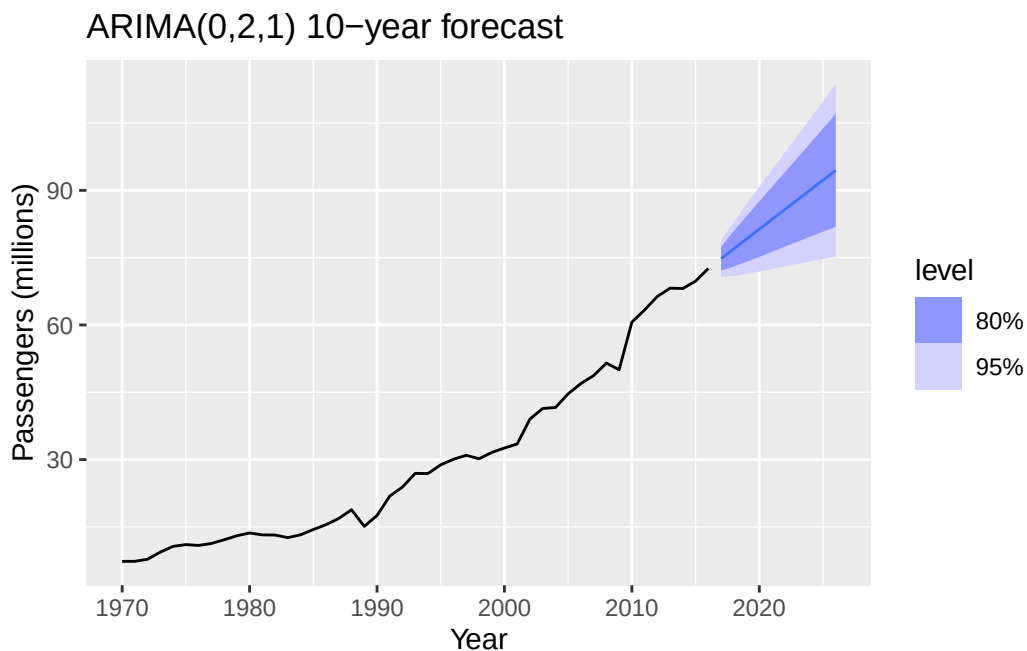


```
fit_auto |> augment() |> features(.innov, ljung_box, lag = 10, dof = 1)
```

```
# A tibble: 1 x 3
  .model lb_stat lb_pvalue
  <chr>    <dbl>    <dbl>
1 arima      6.70      0.669
```

All ACF spikes lie inside the bands and the Ljung–Box test gives $Q^* = 6.70$, $p = 0.669$ — residuals are consistent with white noise.

```
fit_auto |>
  forecast(h = 10) |>
  autoplot(aus_airpassengers) +
  labs(title = "ARIMA(0,2,1) 10-year forecast", y = "Passengers (millions)")
```



Forecasts 2017–2026: 74.8, 77.0, 79.2, 81.3, 83.5, 85.7, 87.9, 90.1, 92.3, 94.5 — nearly linear with slope ≈ 2.2 per year.

b. Backshift operator form.

For ARIMA(0, 2, 1), the model is $(1 - B)^2 X_t = (1 + \theta_1 B) W_t$. Substituting $\hat{\theta}_1 = -0.8963$,

$$(1 - B)^2 X_t = (1 - 0.8963 B) W_t, \quad W_t \sim WN(0, 4.308).$$

c. ARIMA(0, 1, 0) with drift.

```
fit_c <- aus_airpassengers |>
  model(arima = ARIMA(Passengers ~ 1 + pdq(0, 1, 0)))
report(fit_c)
```

Series: Passengers

Model: ARIMA(0,1,0) w/ drift

Coefficients:

constant

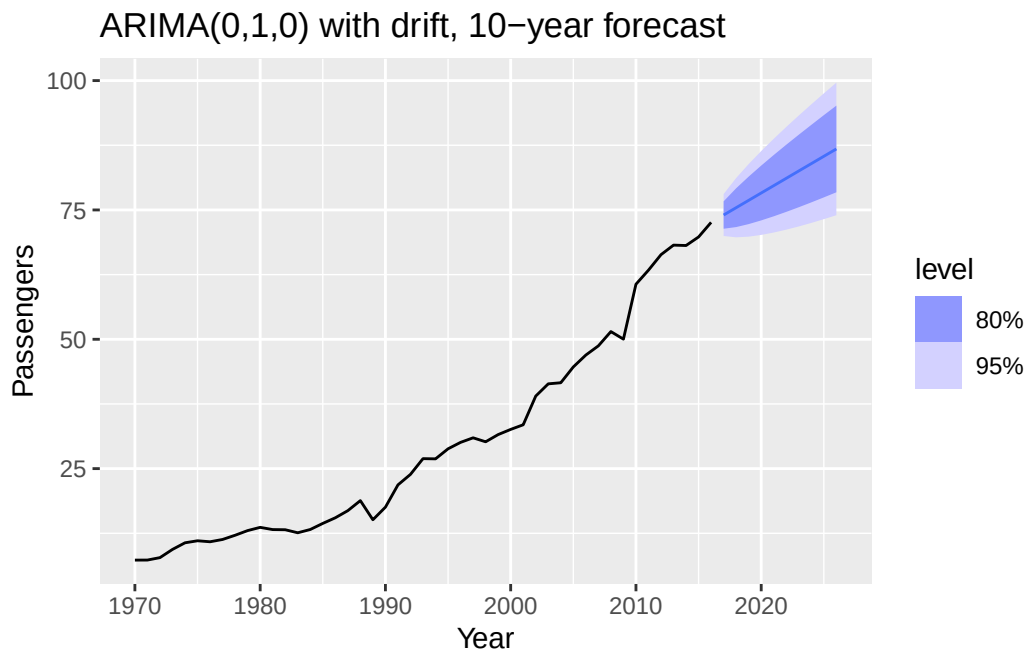
1.4191

s.e. 0.3014

sigma² estimated as 4.271: log likelihood=-98.16

AIC=200.31 AICc=200.59 BIC=203.97

```
fit_c |>
  forecast(h = 10) |>
  autoplot(aus_airpassengers) +
  labs(title = "ARIMA(0,1,0) with drift, 10-year forecast")
```



Random walk with drift $\hat{c} = 1.4191$ (AICc = 200.59). Forecasts 74.0, 75.4, ..., 86.8 — a straight line with slope 1.42, smaller than part (a)'s ≈ 2.2 . Since ARIMA(0,2,1) implies ΔX_t follows

an MA(1), its forecast extrapolates the *recent* change in slope rather than a fixed drift, better matching the steepening growth near the end of the series.

d. **ARIMA(2, 1, 2) with drift.**

```
fit_d <- aus_airpassengers |>
  model(arima = ARIMA(Passengers ~ 1 + pdq(2, 1, 2)))
report(fit_d)
```

Series: Passengers

Model: ARIMA(2,1,2) w/ drift

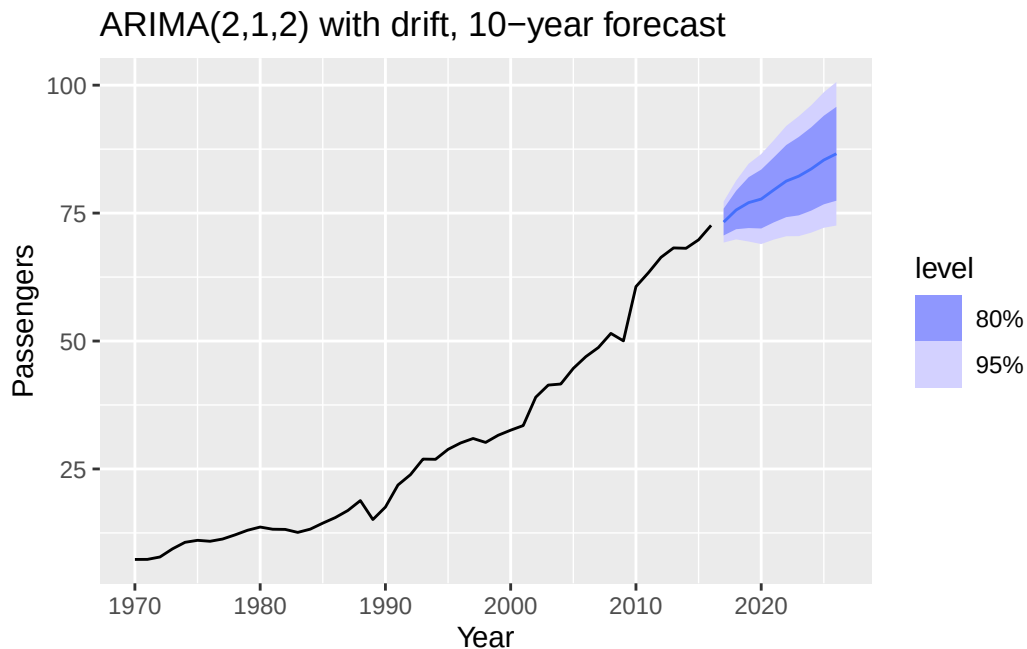
Coefficients:

	ar1	ar2	ma1	ma2	constant
	-0.5518	-0.7327	0.5895	1.0000	3.2216
s.e.	0.1684	0.1224	0.0916	0.1008	0.7224

sigma² estimated as 4.031: log likelihood=-96.23

AIC=204.46 AICc=206.61 BIC=215.43

```
fit_d |>
  forecast(h = 10) |>
  autoplot(aus_airpassengers) +
  labs(title = "ARIMA(2,1,2) with drift, 10-year forecast")
```



AICc = 206.61 (worse than part (a)'s 198.32). Forecasts 73.2, 75.6, 77.1, 77.7, 79.5, ..., 86.6 show a mild wavy pattern from the complex AR roots, tracking slightly below the ARIMA(0,2,1) forecasts.

Removing the constant.

```
fit_d2 <- tryCatch(  
  aus_airpassengers |>  
  model(arima = ARIMA(Passengers ~ 0 + pdq(2, 1, 2))),  
  error = function(e) e  
)  
fit_d2
```

```
# A mable: 1 x 1  
      arima  
    <model>  
1 <NULL model>
```

Without the drift, ARIMA() returns NULL with the message “*non-stationary AR part from CSS*”: the CSS optimiser cannot find a stationary AR, so estimation fails.

e. ARIMA(0, 2, 1) with a constant.

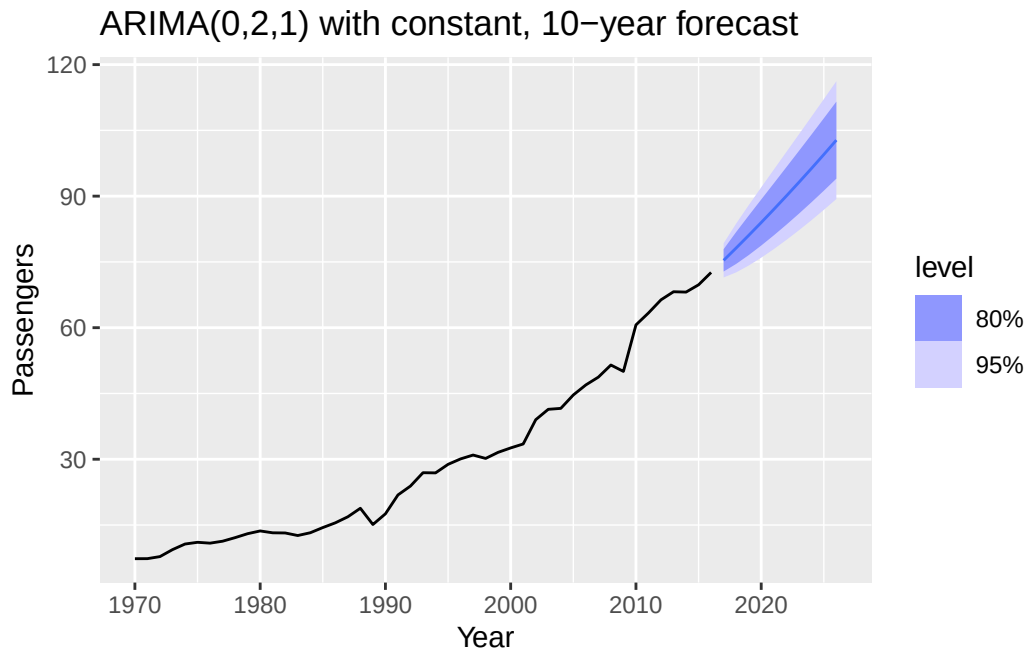
```
fit_e <- aus_airpassengers |>  
  model(arima = ARIMA(Passengers ~ 1 + pdq(0, 2, 1)))  
report(fit_e)
```

```
Series: Passengers  
Model: ARIMA(0,2,1) w/ poly
```

```
Coefficients:  
      ma1  constant  
    -1.0000    0.0571  
s.e.    0.0585    0.0213
```

```
sigma^2 estimated as 3.855:  log likelihood=-95.1  
AIC=196.21  AICc=196.79  BIC=201.63
```

```
fit_e |>  
  forecast(h = 10) |>  
  autoplot(aus_airpassengers) +  
  labs(title = "ARIMA(0,2,1) with constant, 10-year forecast")
```



ARIMA() warns of a polynomial trend. With $d = 2$ and a non-zero constant c , X_t carries a quadratic term $\frac{c}{2}t^2$. Forecasts 75.4, 78.2, ..., 103.0 curve upward (yearly increments grow from 2.8 to 3.3). This quadratic extrapolation is unrealistic; prefer the model in (a).

2. Seasonal ARIMA

a. Load the data.

```
diabetes <- read_rds("../_data/cleaned/diabetes.rds")
head(diabetes)
```

```
# A tsibble: 6 x 3 [1M]
  Month TotalC Cost
  <mth>   <dbl> <dbl>
1 1991 Jul 3526591 3.53
2 1991 Aug 3180891 3.18
3 1991 Sep 3252221 3.25
4 1991 Oct 3611003 3.61
5 1991 Nov 3565869 3.57
6 1991 Dec 4306371 4.31
```

b. Build Y: log + seasonal difference.

```
dbt <- diabetes |> mutate(Y = difference(log(Cost), lag = 12))
```

c. **KPSS test on Y.**

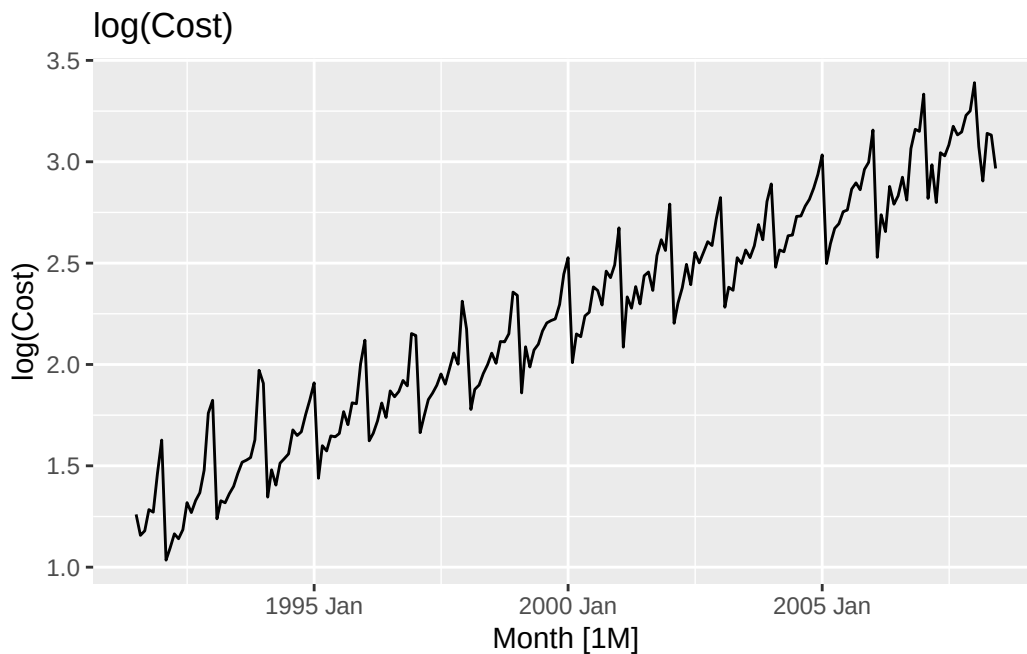
```
dbt |> features(Y, unitroot_kpss)
```

```
# A tibble: 1 x 2
  kpss_stat kpss_pvalue
    <dbl>      <dbl>
1      0.209         0.1
```

KPSS stat = 0.209, p-value = 0.1 (capped at 0.1). We fail to reject stationarity, so Y is consistent with being stationary.

d. **Time plot of log(Cost).**

```
dbt |> autoplot(log(Cost)) +
  labs(title = "log(Cost)")
```



$\log(\text{Cost})$ has a clear upward trend plus annual seasonality. If $\log(\text{Cost}_t) \approx \alpha + \beta t + s_t + u_t$ with period-12 s_t , then $Y_t = 12\beta + (u_t - u_{t-12})$: the trend collapses to the constant 12β and the seasonal component cancels.

e. Fit an ARIMA model to $\log(\text{Cost})$.

```
fit2 <- dbt |> model(arima = ARIMA(log(Cost)))
report(fit2)
```

```
Series: Cost
Model: ARIMA(2,0,2)(1,1,1)[12] w/ drift
Transformation: log(Cost)
```

Coefficients:

	ar1	ar2	ma1	ma2	sar1	sma1	constant
	0.7211	0.1081	-0.6809	0.1198	0.2236	-0.8464	0.0151
s.e.	0.2364	0.2064	0.2316	0.1752	0.1112	0.0835	0.0004

```
sigma^2 estimated as 0.003595: log likelihood=266.01
AIC=-516.02 AICc=-515.24 BIC=-489.96
```

Fitted model: **ARIMA(2,0,2)(1,1,1)₁₂ with drift** ($\hat{\sigma}^2 = 0.00360$, $\text{AICc} = -515.24$).

f. Number of fitted parameters.

```
tidy(fit2)
```

```
# A tibble: 7 x 6
  .model term      estimate std.error statistic p.value
  <chr>  <chr>      <dbl>    <dbl>    <dbl>    <dbl>
1 arima  ar1         0.721    0.236      3.05 2.61e- 3
2 arima  ar2         0.108    0.206      0.524 6.01e- 1
3 arima  ma1        -0.681    0.232     -2.94 3.69e- 3
4 arima  ma2         0.120    0.175      0.684 4.95e- 1
5 arima  sar1         0.224    0.111      2.01 4.56e- 2
6 arima  sma1        -0.846    0.0835    -10.1 1.32e-19
7 arima  constant    0.0151 0.000426    35.4 5.43e-86
```

7 fitted coefficients (ar1, ar2, ma1, ma2, sar1, sma1, constant). Counting σ^2 , $k = 8$, consistent with $\text{AIC} + 2 \log L = -516.02 + 532.02 = 16 = 2k$.

3. Model selection

a. Null hypothesis and assumptions of the ADF test.

H_0 : the series has a **unit root** ($I(1)$); H_1 : stationary (possibly around a trend). The ADF regresses ΔX_t on X_{t-1} and p lagged differences. Assumptions: linear AR data-generating process with mean-zero, finite-variance, (approximately) white-noise innovations; p large enough to whiten residuals; and the included deterministic terms (constant / trend) match those of the data under H_1 .

b. Can ADF and KPSS both have large p-values?

Yes. ADF and KPSS have opposite nulls (unit root vs stationarity), so large p-values mean *failing to reject* their respective nulls. In short or borderline samples (e.g. ϕ close to 1), both tests can have low power, so neither rejects. The conclusion is then that the data are inconclusive — use differencing + re-testing, plots, or domain knowledge.

c. AIC and AICc penalties for the Q2 model.

With $k = 8$ and $n = 204$:

```
n <- nrow(diabetes); k <- 8
c(AIC_penalty = 2 * k,
  AICc_penalty = 2 * k + 2 * k * (k + 1) / (n - k - 1))
```

AIC_penalty	AICc_penalty
16.00000	16.73846

AIC penalty = $2k = 16$; AICc penalty = $16 + 144/195 \approx 16.738$.

d. Fit an ETS model and compare.

```
fit_both <- diabetes |>
  model(
    arima = ARIMA(log(Cost)),
    ets    = ETS(Cost)
  )
glance(fit_both) |> select(.model, log_lik, AIC, AICc, BIC)
```

```
# A tibble: 2 x 5
  .model log_lik  AIC  AICc  BIC
<chr>    <dbl> <dbl> <dbl> <dbl>
1 arima    266. -516. -515. -490.
2 ets     -417.  868.  871.  924.
```

ETS selects **ETS(M, A, M)**. Log-likelihoods and AICc:

Model	log L	AICc
ARIMA on log(Cost)	+266.01	−515.24
ETS(M, A, M) on Cost	−416.97	871.23

e. Can we compare them via log-likelihood or AICc?

No. The two are on different scales ($\log(\text{Cost})$ vs Cost), and the Jacobian shifts the ARIMA log-likelihood by $-\sum_t \log(\text{Cost}_t)$. To compare, fit both on the same scale, adjust the ARIMA log-likelihood, or use time-series cross-validation.

4. Dynamic regression (adapted from Q10.4 in Hyndman & Athanasopoulos)

a. Inflation adjustment.

```
acc <- aus_accommodation |>
  mutate(AdjTakings = Takings / CPI * 100)
```

b. Fit a dynamic regression model for each state.

```
knot <- yearquarter("2008 Q1")
fits <- acc |>
  model(dr = ARIMA(AdjTakings ~ season() + trend(knots = knot)))
fits
```

```
# A mable: 8 x 2
# Key:      State [8]
# State      dr
# <chr>      <model>
1 Australian Capital Territory <LM w/ ARIMA(1,0,0) errors>
2 New South Wales             <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
3 Northern Territory          <LM w/ ARIMA(0,0,1)(1,0,0)[4] errors>
4 Queensland                  <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
5 South Australia              <LM w/ ARIMA(1,0,0)(1,0,0)[4] errors>
6 Tasmania                    <LM w/ ARIMA(0,0,1)(1,0,0)[4] errors>
7 Victoria                    <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
8 Western Australia           <LM w/ ARIMA(1,0,0) errors>
```

Each row shows the ARIMA error structure chosen per state.

c. **Victoria.**

```
vic <- fits |> filter(State == "Victoria")
report(vic)
```

Series: AdjTakings

Model: LM w/ ARIMA(1,0,0)(0,0,1)₄ errors

Coefficients:

	ar1	sma1	season()year2	season()year3	season()year4
	0.6596	0.3756	-57.8760	-36.0249	-12.9360
s.e.	0.0866	0.1242	3.9693	4.6151	4.0447
	trend(knots = knot)trend	trend(knots = knot)trend_41	intercept		
		3.1995	0.3892	264.4398	
s.e.		0.4871	0.9790	13.3769	

sigma² estimated as 170.8: log likelihood=-291.63

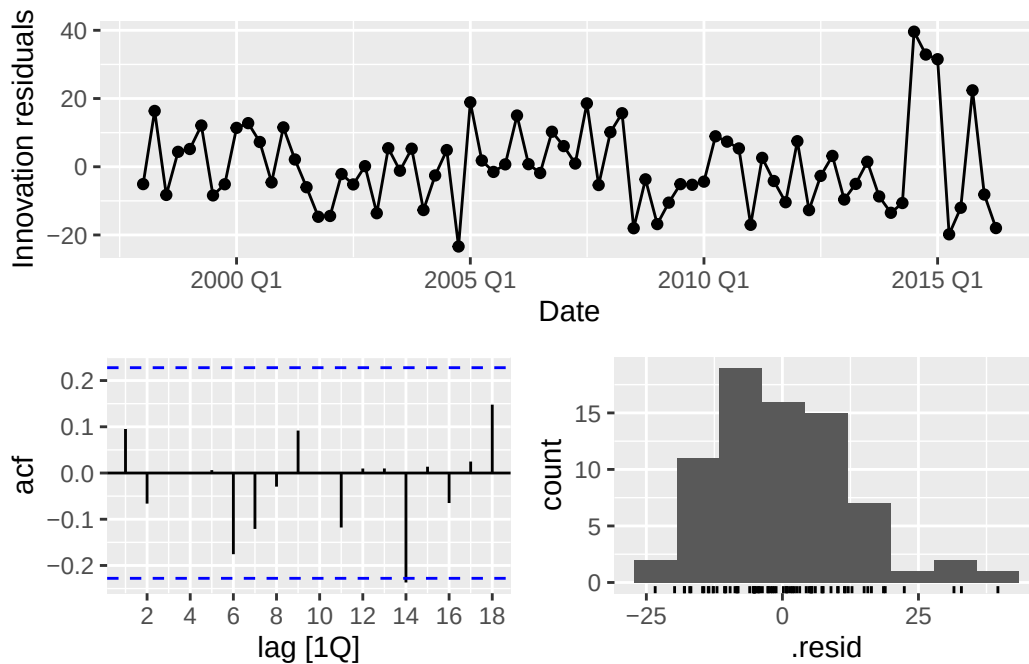
AIC=601.26 AICc=604.07 BIC=622

Victoria: linear regression with **ARIMA(1,0,0)(0,0,1)₄** errors. Intercept = 264.4; seasonal dummies (Q1 reference) year2 = -57.9, year3 = -36.0, year4 = -12.9 (all significant); pre-knot slope = 3.20 (significant); post-knot increment = 0.389 ($p = 0.69$, not significant); AR1 = 0.660, SMA1 = 0.376 ($p = 0.003$).

Constant seasonality? The significant SMA1 shows that **residual seasonal dependence remains after the fixed dummies** — the dummies alone do not fully capture the seasonal structure, and the SMA1 term absorbs the leftover seasonal correlation.

d. **Residual diagnostics for Victoria.**

```
vic |> gg_tsresiduals()
```



```
vic |> augment() |> features(.innov, ljung_box, lag = 8, dof = 0)
```

```
# A tibble: 1 x 4
  State   .model lb_stat lb_pvalue
  <chr>   <chr>   <dbl>   <dbl>
1 Victoria dr         4.90     0.769
```

No ACF spike falls outside the bands; Ljung–Box at lag 8 gives $Q^* = 4.90$, $p = 0.769$. Residuals are consistent with white noise.

e. **Forecast CPI, then forecast Victoria takings to the end of 2017.**

The data ends in 2016 Q2, so we need $h = 6$ quarters (2016 Q3 through 2017 Q4).

```
# Forecast CPI separately with its own ARIMA model
cpi_ts <- acc |>
  filter(State == "Victoria") |>
  select(Date, CPI) |>
  as_tsibble(index = Date)
fit_cpi <- cpi_ts |> model(cpi_arima = ARIMA(CPI))
report(fit_cpi)
```

Series: CPI
 Model: ARIMA(0,1,0) w/ drift

Coefficients:
 constant
 0.5699
 s.e. 0.0520

sigma^2 estimated as 0.2003: log likelihood=-44.37
 AIC=92.74 AICc=92.91 BIC=97.32

```
fcst_cpi <- fit_cpi |> forecast(h = 6)

# Supply forecasted CPI as new_data to the Victoria dynamic regression
new_data_vic <- new_data(acc |> filter(State == "Victoria"), 6) |>
  mutate(CPI = fcst_cpi$.mean)

fcst_adj <- vic |> forecast(new_data = new_data_vic)

takings_fcst <- as_tibble(fcst_adj) |>
  select(Date, AdjTakings_hat = .mean) |>
  mutate(CPI = new_data_vic$CPI,
         Takings_hat = AdjTakings_hat * CPI / 100)
takings_fcst
```

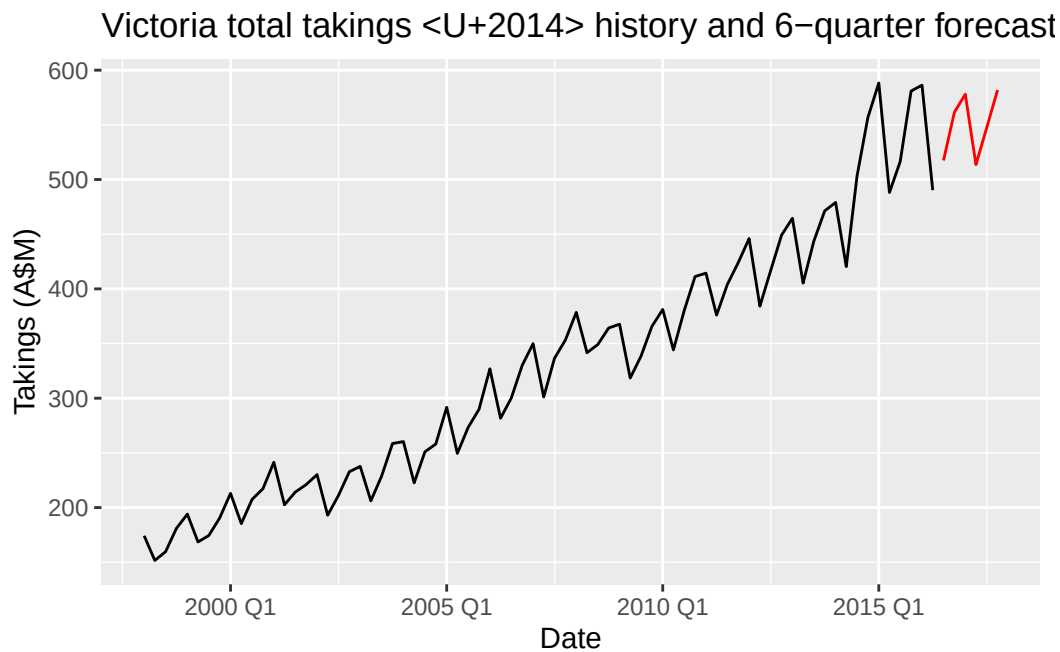
```
# A tibble: 6 x 4
   Date AdjTakings_hat CPI Takings_hat
  <qtr>      <dbl> <dbl>      <dbl>
1 2016 Q3         474.  109.        517.
2 2016 Q4         512.  110.        562.
3 2017 Q1         524.  110.        578.
4 2017 Q2         463.  111.        514.
5 2017 Q3         491.  111.        547.
6 2017 Q4         519.  112.        582.
```

CPI: **ARIMA(0,1,0) with drift** ($\hat{c} = 0.570$ per quarter). Re-inflating via $\widehat{\text{Takings}} = \widehat{\text{AdjTakings}} \cdot \widehat{\text{CPI}}/100$:

Quarter	$\widehat{\text{AdjTakings}}$	$\widehat{\text{CPI}}$	$\widehat{\text{Takings}}$ (A\$M)
2016 Q3	473.99	109.17	517.45

Quarter	$\widehat{\text{AdjTakings}}$	$\widehat{\text{CPI}}$	$\widehat{\text{Takings (A\$M)}}$
2016 Q4	511.67	109.74	561.50
2017 Q1	523.98	110.31	578.01
2017 Q2	463.23	110.88	513.62
2017 Q3	491.15	111.45	547.39
2017 Q4	519.47	112.02	581.90

```
acc |>
  filter(State == "Victoria") |>
  ggplot(aes(x = Date, y = Takings)) +
  geom_line() +
  geom_line(data = takings_fcst, aes(y = Takings_hat), colour = "red") +
  labs(title = "Victoria total takings - history and 6-quarter forecast",
       y = "Takings (A\$M)")
```



Forecast preserves the Q1-peak / Q2-trough pattern and continues the pre-2008 linear rise.

- f. **Sources of uncertainty not reflected in the intervals.** The intervals propagate only the Victoria ARIMA innovation variance. They ignore: (i) CPI forecast uncertainty (point forecasts were plugged in as regressors); (ii) parameter estimation uncertainty in the Victoria model; (iii) model/specification uncertainty (knot at 2008 Q1, piecewise-linear trend, deterministic dummies, ARIMA(1,0,0)(0,0,1)₄ errors).

5. State space models

Consider

$$T_t = T_{t-1} + \epsilon_t, \quad S_t = 0.8 S_{t-4} + \eta_t, \quad Y_t = T_t + S_t + V_t,$$

with $\epsilon_t \sim WN(0, \sigma_1^2)$, $\eta_t \sim WN(0, \sigma_2^2)$, $V_t \sim WN(0, \sigma_3^2)$, mutually uncorrelated.

a. Linear Gaussian state space form.

Since S_t depends on S_{t-4} , keep three lags in the state:

$$\mathbf{X}_t = (T_t, S_t, S_{t-1}, S_{t-2}, S_{t-3})^\top.$$

Then

$$\mathbf{X}_t = F \mathbf{X}_{t-1} + \mathbf{w}_t, \quad Y_t = H \mathbf{X}_t + V_t,$$

with

$$F = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{w}_t = \begin{pmatrix} \epsilon_t \\ \eta_t \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad Q = \text{Cov}(\mathbf{w}_t) = \text{diag}(\sigma_1^2, \sigma_2^2, 0, 0, 0),$$

and

$$H = (1 \quad 1 \quad 0 \quad 0 \quad 0), \quad R = \text{Var}(V_t) = \sigma_3^2.$$

Row 1 gives $T_t = T_{t-1} + \epsilon_t$, row 2 gives $S_t = 0.8 S_{t-4} + \eta_t$ (the 5th entry of $\mathbf{X}_{t-1}$ is S_{t-4}), rows 3–5 shift the seasonal lags, and $H \mathbf{X}_t + V_t = T_t + S_t + V_t = Y_t$.

b. Does $\lim_{t \rightarrow \infty} \text{Var}(S_t - \hat{S}_{t|n})$ exist?

With n fixed and $t \rightarrow \infty$, $\hat{S}_{t|n}$ is a long-horizon forecast. Since $|0.8| < 1$, S_t is stationary with $\gamma_S(0) = \sigma_2^2 / (1 - 0.64) = \sigma_2^2 / 0.36$. As $t \rightarrow \infty$, $\hat{S}_{t|n} \rightarrow 0$, so the limit **exists** and equals

$$\lim_{t \rightarrow \infty} \text{Var}(S_t - \hat{S}_{t|n}) = \frac{\sigma_2^2}{1 - 0.8^2} = \frac{25\sigma_2^2}{9}.$$

c. Does $\lim_{t \rightarrow \infty} \text{Var}(T_t - \hat{T}_{t|n})$ exist?

T_t is a random walk, and for $t > n$, $\hat{T}_{t|n} = \hat{T}_{n|n}$. Hence

$$\text{Var}(T_t - \hat{T}_{t|n}) = \text{Var}(T_n - \hat{T}_{n|n}) + (t - n)\sigma_1^2 \rightarrow \infty.$$

The limit **does not exist** — a random walk's forecast variance grows without bound.